Coventry University
PSB Academy

Automatic Rain-Detecting and Retractable Awning



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A thesis submitted in partial fulfilment of the University's requirements for the Bachelor of Engineering with Honours in Mechanical Engineering

Declaration

The project titled “Automatic Rain-Detecting and Retractable Awing”, is declared that it is researched and done under the supervision of “Dr Lim Chong Lye”, “Head of Mechanical Engineering at Coventry University, PSB Academy”, is the author own work with no plagiarism and to give references to existing works and research.

This project is carried out for Final Year Project with the academic purpose, aiming to create a smart roof system that used Arduino Programming to response the rainfall and act under the given control with the manual override switch. The main goal of the project to achieve the real time world using for the residential houses and other aspect of areas where rain fall is common and unpredictable.

This paper is also ensured that it will only be submitted to Coventry University and has not been submitted to any other institutes.

Name:

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Signature:



Date:
26/04/2025

Acknowledgement

I would like to give a sincere thanks to all the people who helped, guided and supported throughout this project of “Automatic Rain-Detecting and Retractable Awning”. Firstly, my great gratitude would be expressed towards “Dr Lim Chong Lye”, “Head of Mechanical Engineering”, for his valuable lessons, guidance and encouragement to the completion of this project. Also, to “Coventry University, PSB Academy”, I am very grateful for letting me use all the needed facilities to do my project testing and simulations and for providing me the helpful sources required to do the research on various related topic to my project.

Abstract

This report is written to find solution which is a smart awning system, which can detect rain and response to the rainfall in minimum time amount. Concerned by the constant rainfall issue and poor work-life balance in Singapore, this project mainly aims to provide affordable, efficient and reliable awning system that can be installed in residential areas. Based on the problem statement, general methodology includes the flow chart, step by step initiation through design specifications, design concepts and development of the prototype and a small-scale model will be built. At the end of methodology, the results which includes, system response time, awning opening and closing time controlled and monitored by Arduino C++ Programming and stress analysis will be conducted as well. The results obtained satisfy the key objective of minimum amount of time taken by the system in response to rain. Finally, some recommendations on the improvement and future progresses to be made in the future which would benefit the users in the real-world applications in certain areas.

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Chapter 1: Introduction

1.1 Background

In the areas where rain is very common and unpredictable, people find it frustrating to give attention on the nature problem that is concerned with their daily task such as laundry drying. It is obvious to install permanent roof, but it will only be able to function one way and cannot give a clear area when the user wants it to be out. For example, in Singapore, the country that has tropical climate consisting of different level of warm and humid conditions and constant rainfall (Chua et al., 2024).

Fig 1.1 shows the monthly rainfall data in Singapore from year 1991 to 2020. That also shows that Singapore experiences rain very often throughout the year.

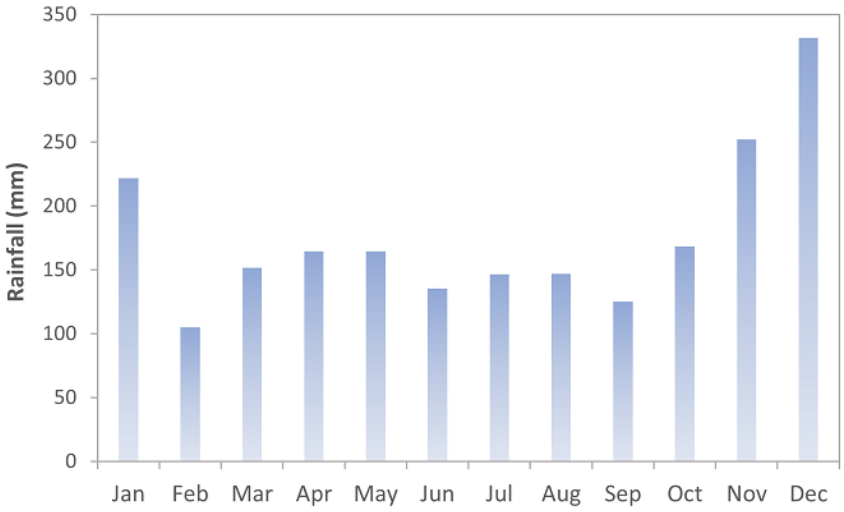


Fig 1.1 Singapore Monthly Rainfall (mm) (1991-2020)

(Meteorological Service Singapore, 2020)



Fig 1.2 Annual Rainfall Distribution in Singapore (1991-2020)

(Meteorological Service Singapore, 2020)

Due to that factor, most of Singapore houses and residential areas have awnings installed in their outdoor areas to prevent from extreme heat and rain. Even though it is functionable like manually fold or retract in a certain way by manpower, it is not optimal to give attention to another task for many people as around **4.02 million** people are working in Singapore (Statista, 2025). Nowadays, lots of automation are available and many countries are starting to adapt into their own smart nation plan and Singapore also has one of its own called "Smart Nation 2.0" (Ministry of Digital Development and Information, 2024). Therefore, the definition of this project is to be part of smart nation plan by using automated smart system that can help public with their daily climate problem.

1.2 Problem Statement

The roof that everyone has nowadays can only provide full coverage and cannot control it willingly. Additionally, if the roof is made controllable manually, it is still inconvenient and unpractical as the user still needs to pay attention to the unpredictable rain. Therefore, some people started asking for a smart roof system for their daily conveniences (Wang et al., 2024). Not only they are asking for the system that can detect rain, fold and unfold panel to get cover, but also, they are looking for the system that is affordable, efficient and reliable. For that reason, this project is to answer their demand by providing Arduino based automatic retractable roof that can be flexible with both user's desire and environment.

1.3 Aim and Objectives

The aim of the project is to design and develop efficient, convenient, smart, cost-effective and safe to use awning which is paired with an automation and IoT based system. It is also to develop smart functions such as automatic rain detection, folding mechanism without manpower and manual override control at will.

The objectives of this project are to –

(1) Conduct literature review for design specifications and generate 3 different concepts based on the literature review

(2) Select design and develop the smart awning system using Arduino controller that detects rainfall and responses by deploying cover for outdoor areas

(3) Develop the system to work properly by responding to the rainfall with minimum time taken

(4) Produce the small-scale model step by step to showcase the key functions of the system

(5) Test the structure of the support through simulations and programming of codes and take notes of the errors

1.4 Scope and Limitations

1.4.1 Scope

Arduino response is very crucial for this kind of project and ensuring it will allow the sensor to detect rain drops and drive the motor to fold and unfold the panels which is the main function of the model. Implementing both automatic sensor and manual override is useful, and it creates a user-friendly interface and allow the system to operate under two different options. Therefore, if one feature fails, the other one will be in standby in case of serious need. The manual mode will be through via switch and app-based control will be considered for future development. Prototype building has been a great help to showcase the expressed ideas and functions. So, it is to build a small-scale working prototype, and it will be able to help demonstrate the basic working principle between the Arduino control unit and the mechanism of retractable awning. The structure of the roof is also very important as it is installed to endure

the strong weather condition such as thunder and rainstorm. As of that, to feature and use materials and structures that can satisfy the mentioned factor.

1.4.2 Limitations

This project is mainly for the residential houses where they have big yards and grounds for some outdoor activities and other. Thus, it is obvious that it cannot be installed in HDB and some apartments as they have tight configurations and regulations. During the development, the model is already scaled to be small scale model just enough to show case the basic function and concept of how the roof panels are controlled with automation, but research will be done on the specifications for the model to be able to use in real time world application. The main focus of this project is about automatic rain sensor that drives the retractable awning. That means that external backup power source and plan, that could be useful for some countries where they do not have constant electricity, will not be considered because of time limit.

1.5 Project Outline

Chapter 1: This chapter introduces the core background concerning with Singapore's climate obtained from Singapore Government's statistics and the explanation on smart nation with the use of automation. Next, the problem that is experiencing in real time world is presented to show how important and beneficial to use automation for daily conveniences (Wang et al., 2024). Aim and objectives also included in this chapter to highlight the aim concerned by the problem statement and the objectives in which the tasks needed for the successful completion of this project. Additionally, the scope that mentions of things that this project will do to achieve its objectives and the limitations that act as the boundary conditions for part of the system or the whole system. The chapter is ended by the Gantt chart that describe the timeline and the processes to be done within that timeline.

Chapter 2: In this chapter, the literature review is provided to strongly back up the field that this project is studying. Two existing research are read through as one targets on the automation processes and its objectives that can be helpful learning for later automation development while another one showcases on the study of the framework for certain roof structure and optimisation. All the research used in literature review are cited in Chapter 2. Later, they both are compared and studied carefully on the parts that are essential and the parts that they miss to do a study.

Chapter 3: This chapter describes on the development of the model. Methodology is initiated by drawing a sketch of a model and its structure. It also includes the part where the sketch will be put onto 3D SolidWorks drawing to be ready for Stress Analysis and other FEA simulations to optimise the best design structure of the system. Arduino programming process is also addressed in this chapter. After that, the selection of the components and the materials that are needed to set up the model is mentioned to bring the project to clear presentation. Finally, the chapter ends with the flow chart that describes the step-by-step methods of the development of the prototype.

Chapter 4: After finished describing the flow chart and development in Chapter 3, Chapter 4 provides the finished prototype and its working performance along with proper discussion and evaluation of the data obtained during Chapter 3. The data includes the performance of the prototype, the time taken by the motors to partially and fully deploy the awning. Since they are to be shown in Chapter 3, Chapter 4 focuses more on the evaluation of the results. Next, trouble shooting of the problem experienced during the tests are to be addressed in Chapter 4.

Chapter 5: This project is concluded with the reflections on work done and discussion on future improvement suggesting on optimal ways to create and develop the better model in the future.

1.6 Gantt Chart

Fig 1.3 below shows the Gantt Chart of the project, and the tasks to be carried out during the timeline of 13 weeks.

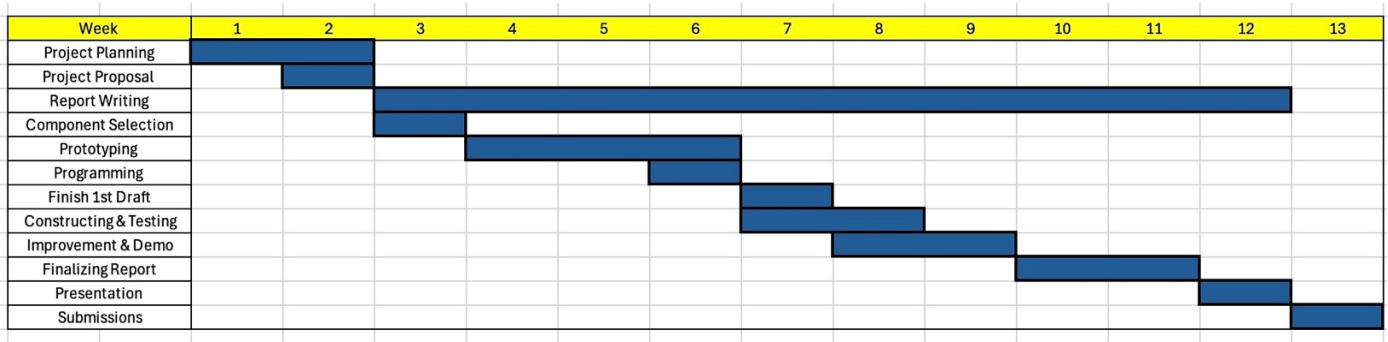


Fig 1.3 Gantt Chart (Made with Excel)

Chapter 2: Literature Review

2.1 Existing Research

Automation and IoT based software usages are becoming more popular in these days as it has been a great help and convenience since day one. Based on the popular automated control system, the rain detection is also considered as the top topic in automation sector. A lot of benefits can be obtained when the rain sensor is integrated with the roof or the awning. According to Muhammad Irsyad Hidayatullah, et al (2023), the roof was made to open and close automatically and to be able to apply for human convenience or environmental matters. They also added the Wi-fi based monitoring with the help of telegram bots to effective viewing of the updates such as weather conditions etc. Moreover, Zhongqi Yu, el al (2024) explains retractable roofs play a huge role in different types of building. She also mentions that ventilation, thermal comfort and energy saving are very important, and it can be managed through careful optimisation of the roof structure and elements. Not only that, but the awning position is also crucial for the reasons based on the article written by Heangwoo Lee, et al (2018). With the proper placement and the protrusion of the awning, it will not only be saving energy but also be an enhancement for the performance of the awning in terms of rain and light. Nevertheless, in terms of durability, diverse weather conditions and the market experience, the topic is very wide and broad which means that every article mentioned has its own strong and weak points.

2.2 Summary

Above 3 articles are found to be very useful to help understand the proposed idea and the purpose of the project as Muhammad Irsyad Hidayatullah, et al (2023) showcase the use of rain sensor for making a roof that is paired with the IoT system for the aviary while Zhongqi Yu, el al (2024) explain and conduct research on the effectiveness of retractable roof in the buildings. Meanwhile, Heangwoo Lee, et al (2018) focus on conducting the integrated awning system with different protrusions and methods. The first 2 articles discuss very well about the important of the smart roof in different areas. The third article provides a clear vision on the nature of lighting environment based on the design. However, the one area that the first article did not discuss is that how the system can also be made for different type of roof structure and different type of purpose. Meanwhile, all the energy, flow and velocity analysis can be found mentioned inside the second article, but it is missing the urban utilization and design development. There is also a room open

for third article in terms of design and automation. It is possible for them to integrate the automation to create a variable angles or protrusion accordingly to the light. Therefore, the key part missing between those three different articles is the design capabilities and structural stress analysis and the market research or the market demand. Thus, that topic will be another future research to be done.

Chapter 3: Methodology

3.1 Introduction

The flow chart which shows the step-by-step process for Chapter 3, Methodology of the project. It is provided below in Fig 3.1.

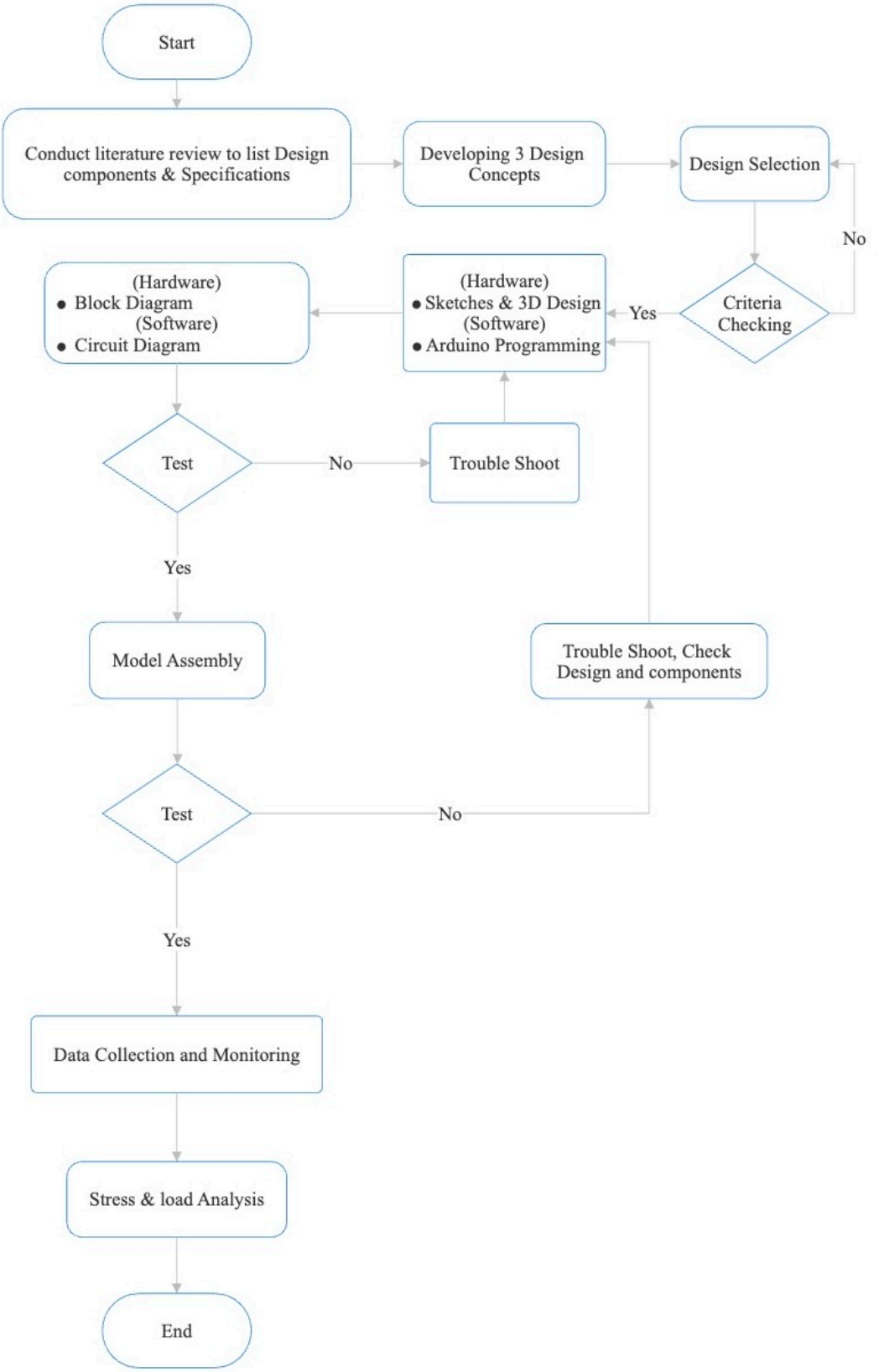


Fig 3.1 Project Methodology Flow Chart

3.2 Design Specifications

To satisfy the objectives mentioned in Chapter 1 and based on the literature review from Chapter 2, design specifications that are required for the successful completion of this project are categorised into 5 different categories. They are as follows –

(1) Functional specifications

(2) Hardware specifications

(3) Software specifications

(4) Environmental specifications

(5) User specifications

3.2.1 Functional Specifications

The main function of the system is to detect rain correctly and give command to the control unit when there is a rainfall. Next is the awning control which is to extend properly with the help of the motors when given signal to the control unit. Everything might seem to be operating precisely as command in most of the time. However, in some cases, the user wants to control it physically. Therefore, manual override is also considered as one of the functional specifications for the system.

3.2.2 Hardware Specifications

The hardware that will be used for this project are listed as follows-

(1) Rain sensor – to detect rain and give command to the motors through control unit when precipitation is present



Fig 3.2 Rain sensor (Hub360, 2025)

(2) Microcontroller (Arduino Uno) – to receive signal from the sensor and drive the motor through programming

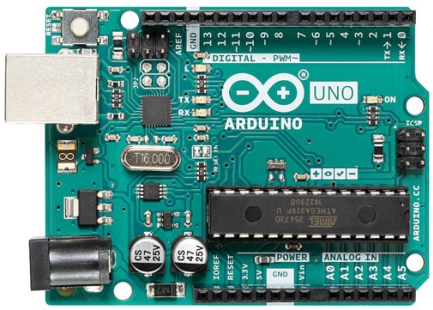


Fig 3.3 Arduino Uno (Element14, 2020)

(3) Servo motors – to retract and extend the arms and the frames of the awning



Fig 3.4 Servo Motor (Xukyo, 2017)

(4) Power supply (5V and External Power supply) – to power the rain sensor directly from Arduino board and power the motors through external power supply with the required voltage to safely run the processes



Fig 3.5 Battery Power Supply (Super Bright Leds, 2015)



Fig 3.6 Cable Power Supply (Robu, 2025)

(5) Housing – to be a cover for the essential modules such as control unit, and batteries

3.2.3 Software Specifications

Arduino microcontroller has been very popular among engineering students in these days as it is easy to program and produce and make small scale projects which is similar to this project. There are several different models of Arduino available. Among them, Arduino UNO is the most used one in the market and it is the best choice for first time robust usage (Arduino, 2025).

The software to program is Arduino IDE which is compatible with lots of Arduino version including Arduino UNO. The program language is mainly **C++**, simply because of many reasons such as easy to code, suitable for automation and IoT based applications (Chatterjee, 2024).

3.2.4 Environmental Specifications

Under normal conditions, Arduino boards can operate very well. Nevertheless, in some cases, when there is a requirement for the operations regarding to weather, it has some limitations and range to operate safely.

The first one is the operation temperature. The normal operating temperature is set to be **from -40°C to +85°C**. All the temperature exceeding that range will affect the board and the components and even damage them permanently. It is also safe to note down that the safest and recommended range by Arduino is **from -25°C to +70°C** (Arduino Support, 2025)**.**

Next is the requirement for servo motors. Most of the servo motors cannot withstand the water. Therefore, it is to install the cover and put the motors under the cover for more durability and longer motor life.

3.2.5 User Specifications

According to the problem statement, users want the system to be convenient. To achieve it, the best option is to add manual override. Thus, the users can have options to switch between automatic system and manual system. Another specification is the user friendliness. This can be achieved by approaching with the system that is easy to install and maintain (Ministry of Digital Development and Information, 2024).

3.3 Design Concepts

3.3.1 Existing Market Research

To be able to develop the possible and potential design concepts for this project, the market research is also made to assist the concepts development. This section will provide 3 market research product that already exist in the market. They are described in Fig 3.7, 3.8 and 2.9 below.



Fig 3.7: Freestanding Awning (Roche Awnings, n.d.)



Fig 3.8: Retractable Canopy (Lanai-outdoor, 2016)



Fig 3.9: Convertible Car Roof (Radu, 2014)

3.3.2 Initial Concepts Development

Concepts 1 (Automatic Rain Detecting and Retractable Awning, Semi-Circle shape)

Much like “Freestanding Awning”, this particular design, shown in both Fig 3.10 and 3.11, is driven by the motor along the designed path to form a semi-circle awning. It is similar to those retractable and foldable fans that people used to carry around in their bags. Since, there are 7 awning blades in total attached by the arm for each blade and vinyl or fabric sheets to fill up

the gap, the motor is the key source to open and close the awning when the rainfall is detected. Furthermore, 3 advantages and disadvantages of this awning will be discussed below.

3 advantages of this system are-

(1) Modernized and Innovative design

(2) Little motor is needed to operate, uses little energy

(3) Fit to install in most of the areas

3 disadvantages of this system are-

(1) Complicated to install

(2) Hard to maintain

(3) Expensive to install

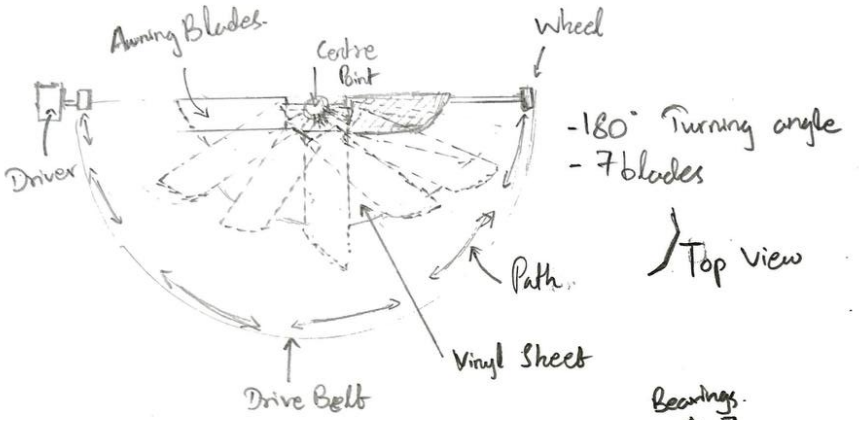


Fig 3.10: Concept 1 (Top View)

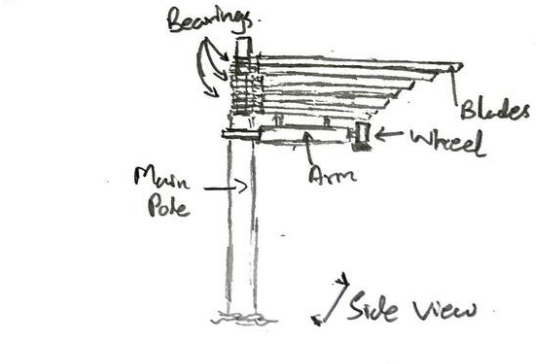


Fig 3.11: Concept 1 (Side View)

Concept 2 (Automatic Rain Detecting and Retractable Awning, Curtain)

Unlike from concept 1, it acts as a automatic curtain that can response to the rainfall according to Fig 3.12 and 3.13. It is very simple as it contains 4 motors mounted in pairs, 2 for each side to open and close the vinyl sheet. When the rainfall is detected, the first set of motors will pull the string made of steel wire that is attached to the very end of the curtain. To close the curtain by the time it stopped raining, the second set of motors from the opposite side will roll the curtain backwards, opening the shade. However, there are still 3 advantages and disadvantages for this design, and they will be addressed below.

3 advantages of this system are-

(1) Simple and Easy to install
(2) Cheap and Efficient
(3) Easy to maintain

3 disadvantages of this system are-

(1) Less Durable for storms
(2) Takes up space
(3) Needs frequent maintenance

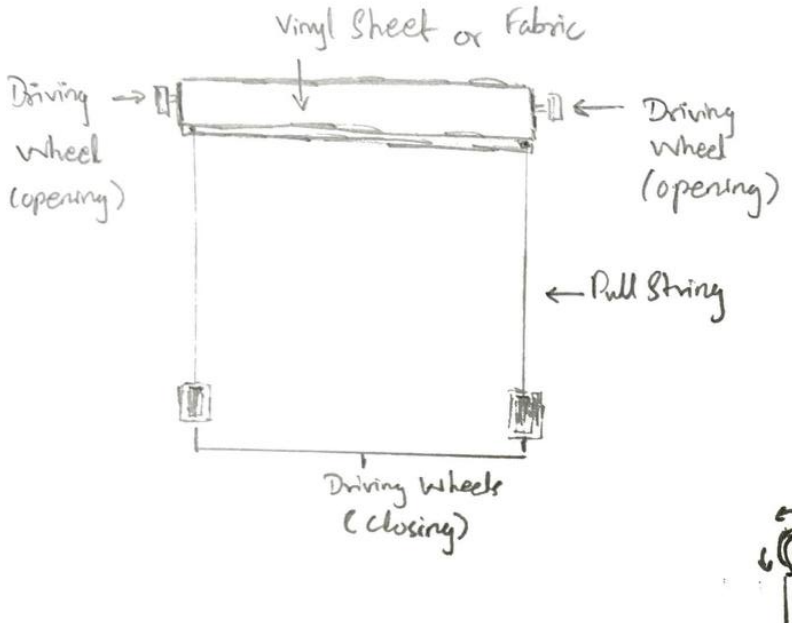


Fig 3.12: Concept 2 (Top View)

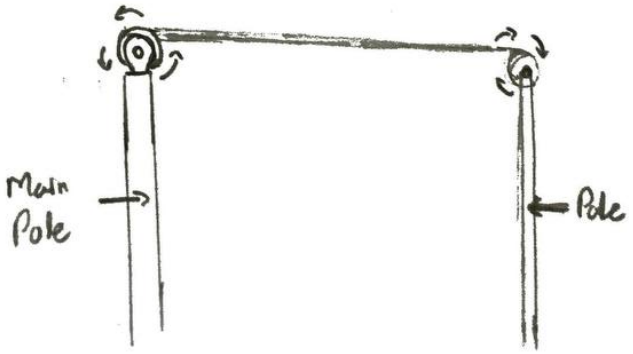


Fig 3.13: Concept 2 (Side View)

Concept 3 (Automatic Rain Detecting and Retractable Awning, Convertible)

Main inspiration taken from the convertible roof of a car, this design, provided in Fig 3.14 and 3.15, focuses on the stress during the weather condition and act as a rainfall responder just like 2 other concepts. Powered by 4 different motors, it comes in 2 different sections, the main panel and the 2nd panel, 2 motors for each section. It also comes with the safe house which is to be the cover for the electrical system and gives the clean look after the rain. Therefore, 3 advantages and disadvantages of this awning system is written below.

3 advantages of this system are-

(1) Durable to most of the storm
(2) Easy to maintain with adjustable settings
(3) Efficiently fashionable

3 disadvantages of this system are-

(1) Need time to time replacement of the motors
(2) Not suitable for tight spaces
(3) Cannot withstand severe weather conditions

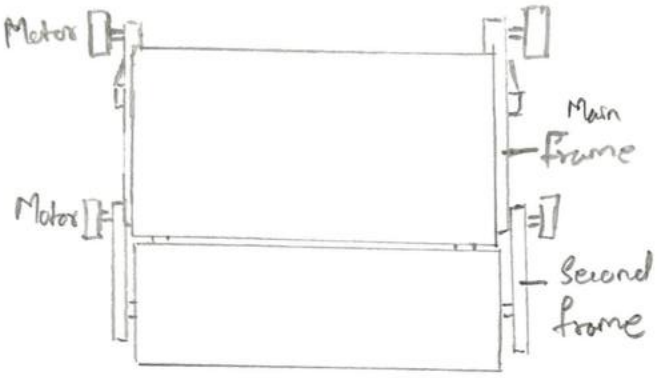


Fig 3.14: Concept 3 (Top View)

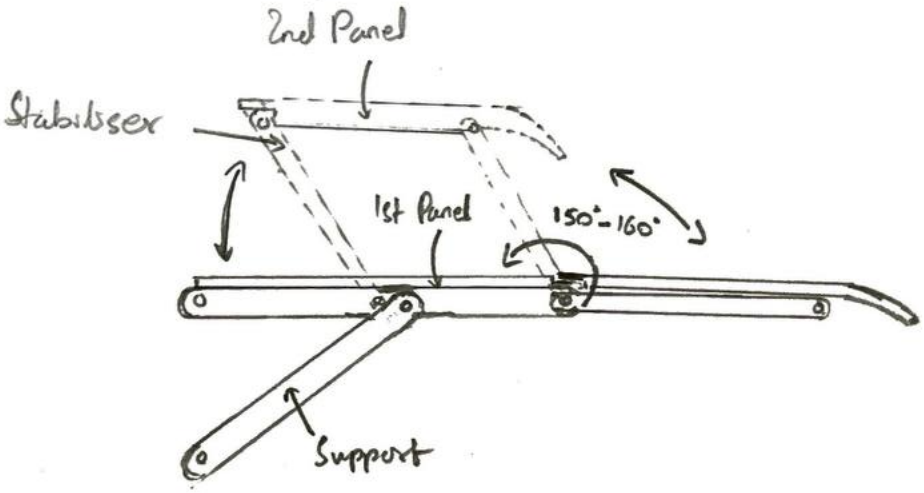


Fig 3.15: Concept 3 (Side View)

3.4 Design Selection

Table 3.1 provides the criteria rating for each concept and the concept that has been chosen based on the required criteria for this project.

		Concept 1		Concept 2		Concept 3	
Criteria	Weighting	Rating (1-10)	Weighted	Rating (1-10)	Weighted	Rating (1-10)	Weighted
Functionability	20%	8	1.6	8	1.6	9	1.8
Efficiency	20%	5	1	4	0.8	7	1.4
Feasibility	20%	5	1	6	1.2	6	1.2
Cost	20%	3	0.6	7	1.4	6	1.2
Ease of Maintenance	10%	2	0.2	7	0.7	7	0.7
Aesthetic	10%	10	1	2	0.2	6	0.6
Total Score	100%	33	3.4/10	34	5.9/10	41	6.9/10
Final Concept			✗		✗		✓

Table 3.1 Design Selection Criteria Marking Table

With the rating score of 41 out of 60 and the weightage score of 6.9 out of 10, **Concept 3** is chosen as the perfect design concept for this project.

3.5 Prototype Fabrication

3.5.1 Materials and Components

Materials

Although the prototype is limited to small-scale project, the materials that are used for the system are determined in large-scale project. Components on the other hand, are listed by the actual components used to operate the small-scale model as the motors, the control system and other electrical specifications with the same concept can scale up depending on the size of the real-life project.

In awning structures, it is categorized into three different sections such as frames, roof, and housing. The materials used for each section will be described below.

Parts	Material used
Frames - Rear bow (left, right) - Front bow (left, right) - Stabiliser (left, right) - Support arm (left, right)	- 6061 Aluminium alloy - 6005 Aluminium alloy - 6063 Aluminium alloy - 6061 Aluminium alloy
Roof - Panel 1 - Panel 2	- Zincalume (Mathews, 2023) - Zincalume (Mathews, 2023)
Housing	- Cast Iron

Table 3.2 List of parts and materials used

For the frames, different types of aluminium alloy have been used for lightweight, and durability while also having a great strength from the alloys. 6061 Aluminium alloy is used for both 1st arm and support arm because the particular alloy possesses the tensile strength of 310 MPa which makes it suitable for the specific area where a lot of force must be exerted (AZO, 2012).

Meanwhile, 6005 Aluminium alloy has been applied to 2nd arm because the alloy has the tensile strength of 270 MPa and it is cheaper than 6061 Aluminium alloy (AZO, 2005). On the other hand, the material for the stabilizer has been given as 6063 Aluminium alloy as it has the lowest tensile strength out of three of them, only having 130 MPa and it is also the cheapest alloy compared to the other two which is the perfect suit for task with small load (AZO, 2005b).

Zincalume is a very popular coating steel used in many constructions because of its properties. It has one of the best corrosion resistance and longevity (MSF, 2024). Therefore, it is chosen as a roofing material for this project instead of galvanised steel which is cheaper. However, for longer life span with better durability, zincalume is still the best option to consider for this project.

Due to the strong durability and the degree of corrosion resistance, cast iron is used for the housing of the general components as it has the tensile strength of 410-560 MPa (Kępczak & Pawlowski, 2013).

Components

The list of electrical components that will be used for this project is listed as follows –

(1) Arduino UNO board x 1
(2) Rain sensor module x 1
(3) Servo motors x 4
(4) Servo driver (PCA9685) x 1
(5) Switch x 1
(6) Capacitors x 4
(7) Jumper wires
(8) 5V 5A external cable supply x 1
(9) Breadboard x 1

3.5.2 Assembly Process

Before the actual assembly of the components and the model, the 3D model of the assembly and the block diagram to show the process flow within the electrical circuit of the components will be provided below.

Concept Assembly (3D model)

- Drawn and designed in real life scale
- Table 3.3 provides dimensions

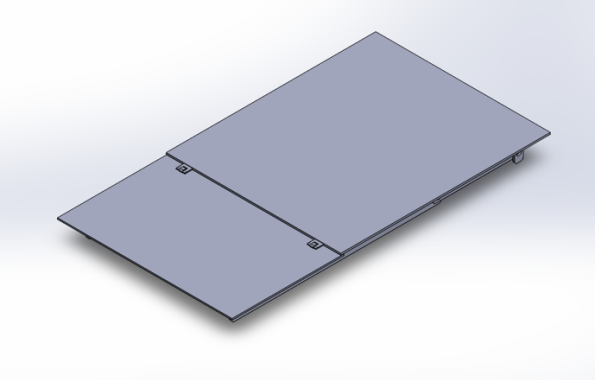


Fig 3.16: 3D View (Top)

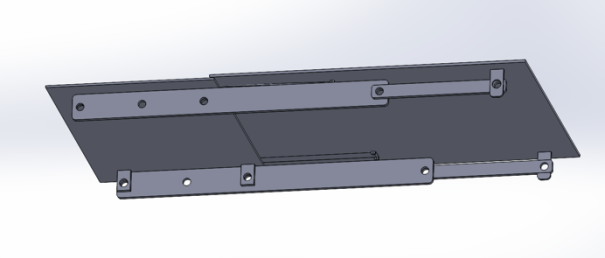


Fig 3.17: 3D View (Bottom)

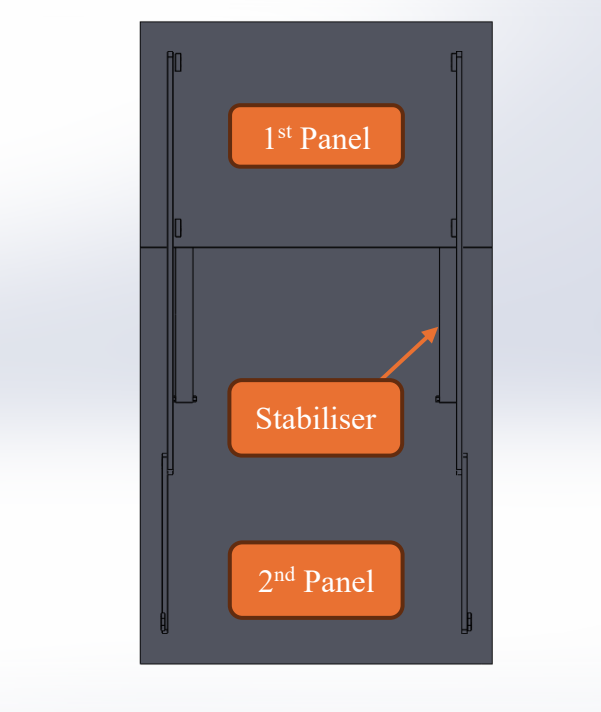


Fig 3.18: Bottom View

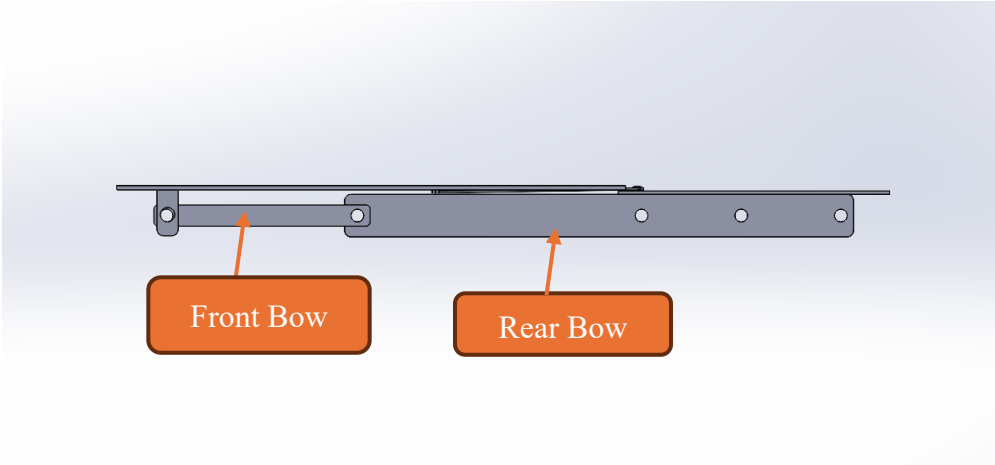


Fig 3.19: Side View

Components	Length (mm)	Width (mm)	Thickness (mm)
Rear Bow	1000	100	15
Front Bow	600	50	15
Stabiliser	500	50	5
Support Arm	583.1	50	8
1st Roof Panel	640	1000	8
2nd Roof Panel	1200	1000	10

Table 3.3: Dimensions of the awning structure

Hole diameter – 30 mm

Block Diagram

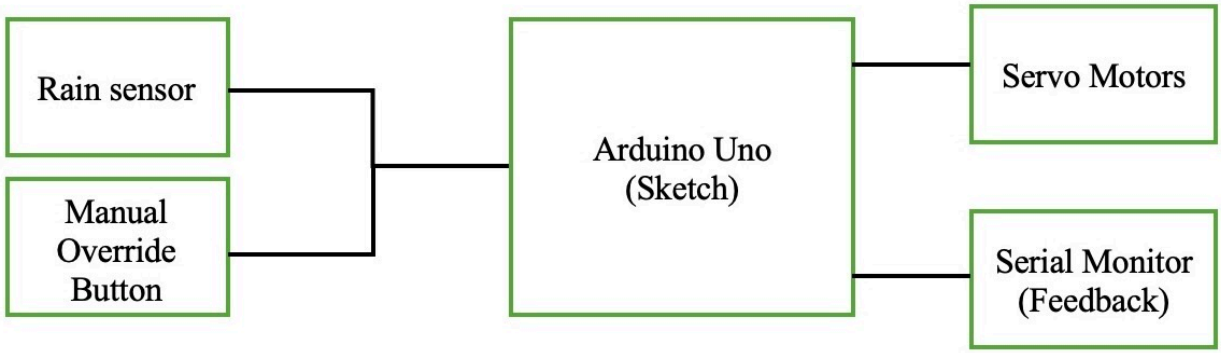


Fig 3.20: Simple Process Block Diagram

Electrical Assembly

By using the electrical components described above, a functional circuit is set up and the steps will be written as follows-

(1) Rain sensor is connected with the analogue module with positive and negative wire and the analogue module is planted on the breadboard

(2) The analogue module is connected with Arduino UNO board with 3 different wires, VCC that is pinned to the 5V power supply that is connected to the breadboard, came from the Arduino board, GND that is connected to the ground of the breadboard which has the ground pin that is connected to ground of the Arduino board, and D0 that is the digital output pin pinned to the digital pin 2 which is declared inside the code.

(3) The switch is installed using input pull up function on the breadboard that allows the switch to operate without the need for resistor and two pins are connected to digital pin

from the control board and the ground from the breadboard at two opposite ends of the switch.

(4) 4 capacitors that has 16V 470 µF of capacitance are placed side by side across the breadboard with proper and same positive and negative placement which will later smooth out the voltage surges and spikes that can occur jitters in servo motors.

(5) 4 servo motors are connected with the capacitors one by one on the breadboard with separated wires, positive and negative accordingly to each of the capacitor's placements.

(6) The fully paired motors with capacitors are then attached to the pin 0,1,2,3 of PCA9865 servo driver one by one accordingly from positive to negative. Additionally, the digital pins of each motor are joined directly to the digital pins from servo driver.

(7) PCA9685 is later paired with Arduino board through 4 different wires, GND to the ground SCL to the analogue pin 5, SDA to the analogue pin 4 and VCC to 5V power supply of the board. The driver is then powered using 5V 5A external power cable through the input, positive and negative ports that is joined by the wires and barrel jack adaptor from the cable.

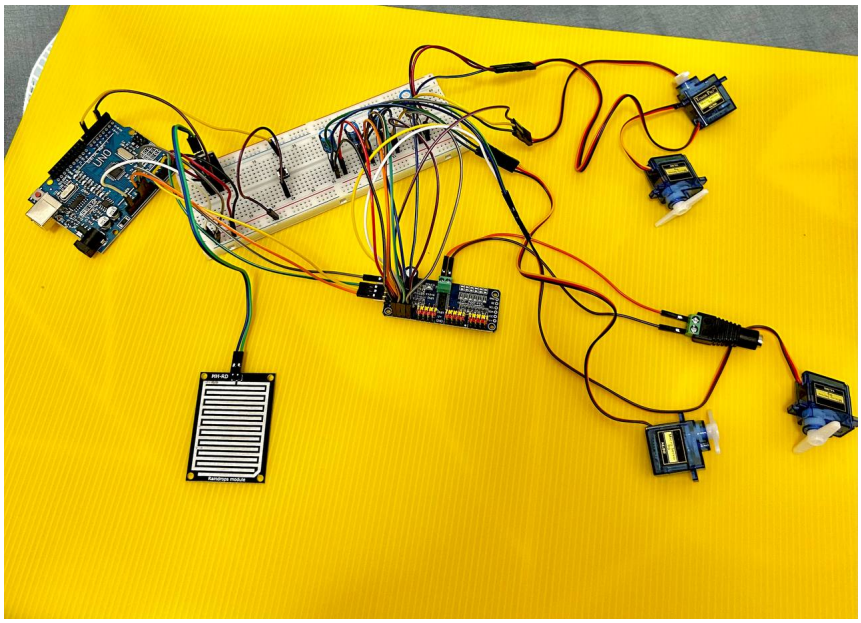


Fig 3.21 Electrical Set up of the model

Frame structure assembly

By using the components listed in Table 3.3, the awning structure is set up according to the steps below-

(1) 2 rear bows are aligned parallelly so that 2 front bows could be connected at each other end of each rear bow.

(2) 1st roof panel is installed onto rear bow with connectors.

(3) Stabilisers are initially joined with 1st roof panel through the connection links located at opposite ends and then joined with the 2nd roof panel to cancel out unnecessary motions.

(4) 2nd roof panel which was paired with the stabilisers is put onto the front bow which will then be linked together with front bow by connectors at opposite ends.

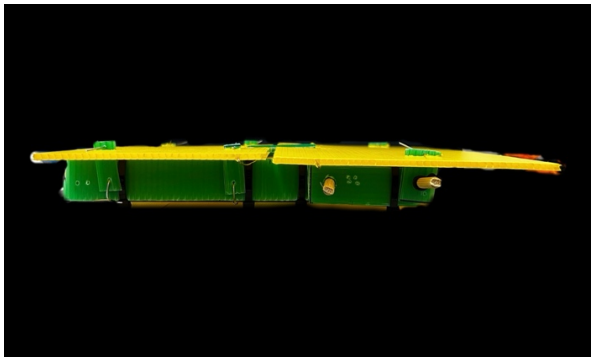


Fig 3.22: Frame Assembly (Side view)

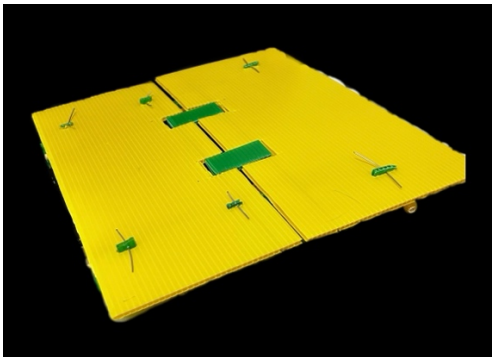


Fig 3.23: Frame Assembly (Top view)

Prototype assembly

After finished assembling the frame of the awning, the whole prototype is also assembled. This process undergoes how the frame is attached to the motors and mounted on the housing. The steps of the assembly are as follows-

(1) The simple housing, described in Fig 3.24, is assembled with the mounting area 164 mm off the ground with 30-degree downward slope to reduce the project of the awning in order to fit with the regulations. Also, two supporting pillars are installed to keep the frame from falling off the mounting.

(2) Rear bow of the frame is paired with two servo motors, servo 1 and servo 2 from left to right respectively. Fig 3.25 shows how it is attached.

(3) Front bow of the frame is attached with another two servo motors, servo 3 and 4 the same way as the process for rear bow from left to right respectively with servo 4 mirrored to servo 3.

(4) Electrical assembly is put inside the housing with the rain sensor set up beside it.

(5) Fig 3.26 and Fig 3.27 show the completely assembled prototype which is ready to be tested for certain criteria.

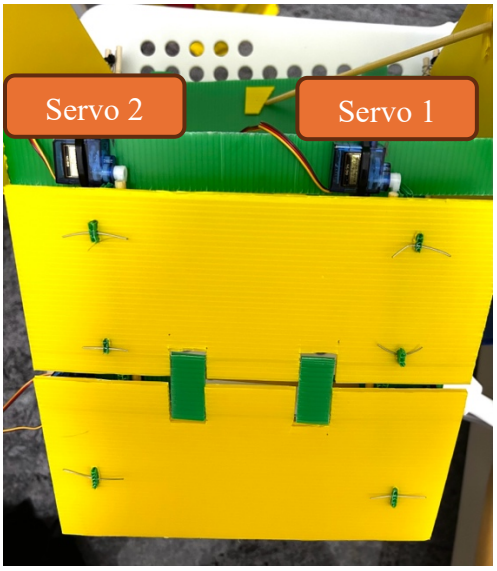


Fig 3.24: Servo mounting with Rear bow



Fig 3.25: Housing

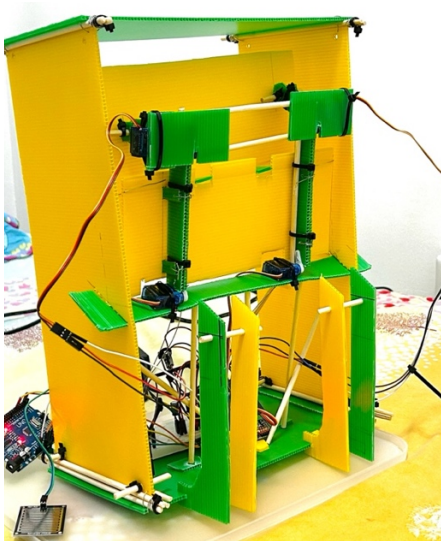


Fig 3.26: Assembled Prototype (Roof Closed)

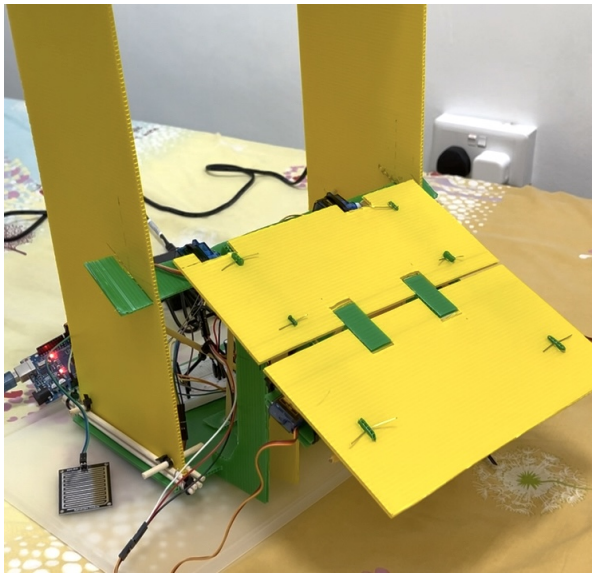


Fig 3.27: Assembled Prototype (Roof Opened)

3.5.3 Testing Environment setup

In normal room temperature, the prototype is placed on the dry place and just using water drops, the roof will be activated, and it can be tested several by dropping water droplets onto the rain sensor to deploy or extract awning and wipe the water away to retract awning. For manual mode, the switch can be pressed first time to open awning and pressed another time to close awning. Key thing to note when using manual mode is that no sudden presses are allowed as it would only let the button edge to be detected and not trigger the signal.

3.6 Monitoring

By opening "Serial Motor" in Arduino IDE through Tools tab, the basic opening and closing can be monitored step by step.

Firstly, the auto mode triggered by the water droplets on the rain is monitor and Fig 3.28 shows it. When the water is dropped on the rain sensor, it gives the signal to the Arduino board which will operate the servo motors, and that operation is shown as "Auto: Rain detected – Opening roof". That information is given out as the user programmed using "**Serial.println**" in the sketch and it can be changed to any information form that the user wants. It is also the same process for the reverse process. When the water is wiped off the sensor, the information, "Auto: No rain- Closing Roof", pops up to present that the roof panels are closing. Then, it will repeat the same information based on numbers of operations the user carries out.

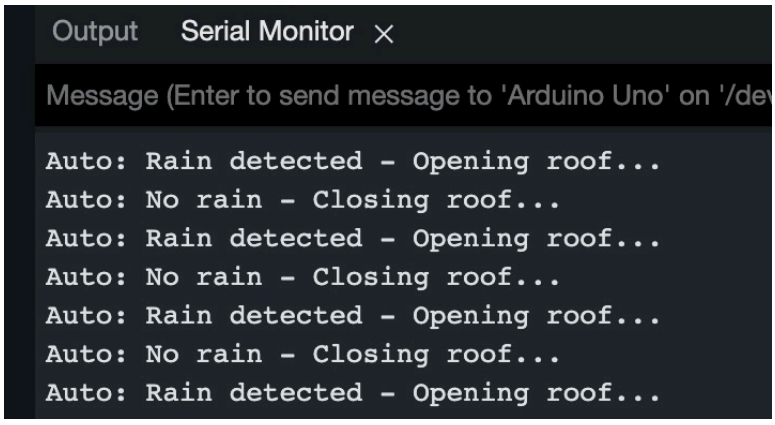


Fig 3.28: Monitoring of Auto mode functions

In Fig 3.29, a switch between auto and manual mode can be monitored. As soon as the button is pressed 1-2 seconds, the information, "Button edge detected" and "Confirmed button press", will show up to allow the manual mode to proceed and give command to the servo motors.

That extra information is to debug the sudden digital debounces in each press. Then, the opening of the roof can be known by monitoring the serial monitor which presents “Manual Override – Opening roof”. Again, when the button is pressed 1-2 seconds for a second time, “Manual Override – Closing roof” will show in monitor, saying the roof panels are closing.

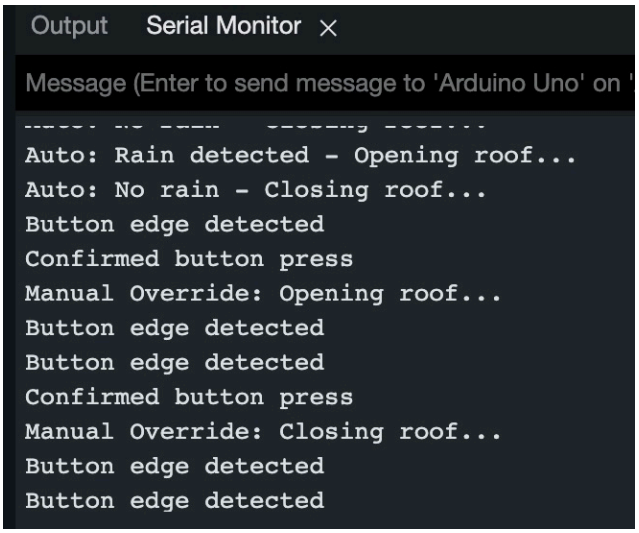


Fig 3.29: Monitoring of Switch between two modes and Manual mode functions

3.7 Stress Analysis

FEA stress analysis of each component of the awning structure is carried out with consideration of wind force. Therefore, there are two forces acting on the structure They are the awning weight from above and the wind force from the side. For maximum conditions, the weather is considered as stormy, and the air speed is assumed to be 25 m/s (Agrox, 2025).

Awning weight = 15 kg x 10 m/s^2 = **150 N**

To calculate the wind force or load, the formula is derived as follows-

$$F = \frac{1}{2} . \rho . C_d . A . V^2$$

where,

F = Wind force

ρ = Air density (1.293 kg/m^3) (Earthdata NASA, n.d.)

C_d= Drag coefficient on flat surfaces (1.28) (NASA, n.d.)

A = Area of the subject that the wind blowing through

V = Air speed

Therefore, the area is taken from the side of the awning structure as the wind is assumed to be mostly blowing from the side and lifting the awning off.

Area, A = 168 m^2

Hence, the wind force, F is obtained as **86889.6 N.**

3.8 Safety Considerations

There are various groups and organizations that help regulate buildings including awnings for public safety in Singapore. Those organizations are listed as follows –

(1) Building and Construction Authority (BCA)
(2) Urban Redevelopment Authority (URA)
(3) Singapore Civil Defence Force (SCDF)

Every building and structure must follow the laws and regulations as directed by those authorities. Thus, this project is also to comply with the safety standards which will be described and discussed below. There are 5 main regulations to note down when developing the awning. They are-

(1) Max protrusion or projection limits (Uni Shades, 2025)
(2) Restrictions on materials usage (Uni Shades, 2025)
(3) Drainage placement (Uni Shades, 2025)
(4) Restrictions on designs (Uni Shades, 2025)

To achieve full users' satisfaction, safety and the approval from BCA, URA and MCST, this project is in full considerations of the regulations of different sectors such as materials, engineering and installation procedures as already mentioned above.

The model is designed for the private properties and not for the common housings or HDB. However, the model is still in consideration of getting approval from URA. URA states that for awning without vertical support, the projection should not exceed 1.4 m into road buffer (Urban Redevelopment Authority, 2025). Therefore, the awning structure is to set up at the downward slope of approximately 29 degrees to achieve the clearances.

SCDF does not actually state any restrictions on what type of material to use but it is to use high tensile strength materials for user safety and maximum efficiency during operation. There is no specific drainage line designed in the roof of the awning since the structure is pitched down for all the water and other debris to flow down.

The installation of the awning structure is as mentioned above in the assembly part. The model is designed and developed to easily assemble and disassemble to comply with installation regulation and design restrictions guided by MCST.

3.9 Summary

Throughout the development process of this project, this report can provide the flow chart for better view of the workflow and tasks. Before initiation of the development processes, the design specifications are listed down into 5 different categories to research for the market to take inspirations that are well-fitted for the prototype. After that, 3 different design concepts are presented with advantages and disadvantages to assist the design criteria marking. Among them, design concept 3 is chosen to be the perfect fit to implement the proposed system. Moreover, the lists of materials required for the structure and the components needed to do the electrical setup are made. Before assembling the prototype, 3D concept model is developed using SolidWorks to provide clear visual on both the possible outcome of the prototype and the reference for real world use. Finally, the prototype is assembled through electrically, and structurally and later tested for response time and the operating time to monitor the performance of the system. Additionally, stress analysis on key parts of structure such as rear bow, front bow and roof panels is carried out to further clarify the durability and tolerance of the awning.

Chapter 4: Results and Discussion

4.1 Test Results

Functional testing (Response Time)

Without the panels and structure attached to the motors, it is weight free to test the response time by the motor under certain signal receiving. The test is carried out 3 times for auto mode to get a better comparison between two options. The average sensor response time is around 0.5 to 0.7 s. The test results are shown in Table 4.1.

Functions (Auto)	Test 1	Test 2	Test 3	Average
Opening roof	1.68 s	1 s	1.95 s	1.54 s
Closing roof	3.2 s	2.7 s	2.88 s	2.93 s
Total time	4.88 s	3.7 s	4.83 s	4.47 s

Table 4.1: Motor Response Time (Auto Mode)

Functional testing (Operating time)

In this part, with the completed assembly, 3 tests are carried out to determine the average operation time taken by the model under the weight from the awning structure and the average sensor response time of around 0.5 to 0.7 s. The test results are shown in Table 4.2.

Functions (Auto)	Test 1	Test 2	Test 3	Average
Opening roof	3.36 s	3.45 s	2.54 s	3.12 s
Closing roof	3.24 s	2.21 s	2.59 s	2.68 s
Total time	6.6 s	5.66 s	5.13 s	5.8 s

Table 4.2: Operating Time (Auto Mode)

By Table 4.1, it is certain that the rain sensor is very responsive as the motors are quick enough to response under 2 s for opening roof and 3 s for closing roof. The average sensor response time when the water droplet is present is around 0.5 to 0.7 s. However, the variations of time in different tests can be due to 3 main reasons. They are-

(1) Difference in rain sensitivity adjustment – low sensitivity would not trigger immediately (DF Robot, n.d.)

(2) Digital Debounces – it is to filter out unwanted input noised from the sensor which may cause some delays while putting on droplets and wiping the droplets away (Wright, 2022)

(3) Human reaction time – the time taken can be varied between 1 to 2 second depending on different person and it can sometimes cause 0.2 seconds of extra time in the stopwatch (Carson, 2024)

It is to say human reaction time does not really matter but the system is not coded with time module and human must use stopwatch to note down time. Thus, it became one of the factors for such differences in time.

Table 4.2 provides the operating time by the completed prototype meaning there is a load exerting on the motors. Nevertheless, the system still operates relatively fast with 3.12 s to fully extract roof and 2.93 s to fully retract roof. Due to reason 2 mentioned above, the rain sensor might be sometime slow to response when doing many tests. That very much explains why the opening time is slower than the closing time. However, the system manages to operate and response to the presence and the absence of the rain under short amount of time.

4.2 Stress Analysis Results

Two different analyses are done. The first analysis uses both the awning weight, and the wind force is added onto rear bow and front bow as the frames are the bases of the structure and considered to be the main part of the whole awning. For the second analysis, only the awning weight is added on top of roof panels and stabiliser as the side areas of each component are very small to determine and less likely that the wind would exert any significant force.

Stress analysis of each part except support arm is shown from Fig 4.1 to 4.10, and they are evaluated after the figures.

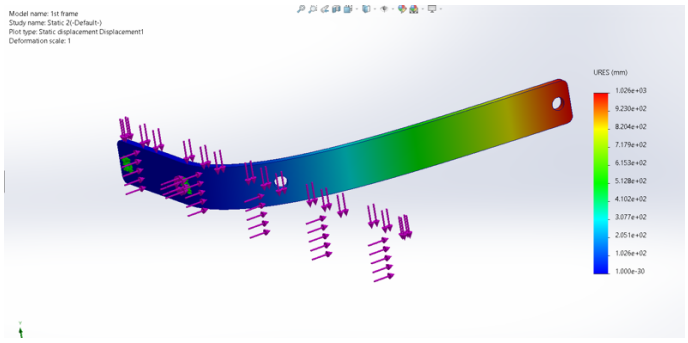


Fig 4.1: Displacement (Rear Bow)

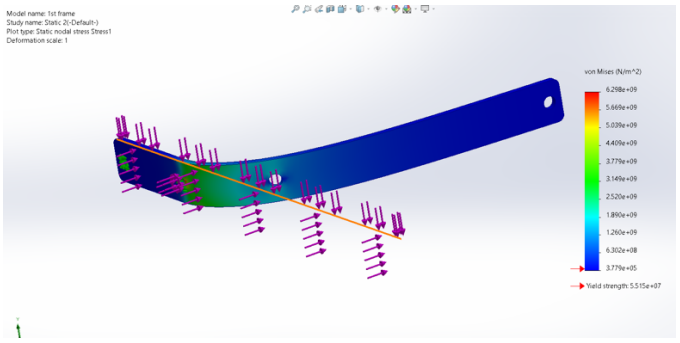


Fig 4.2: Stress (Rear Bow)

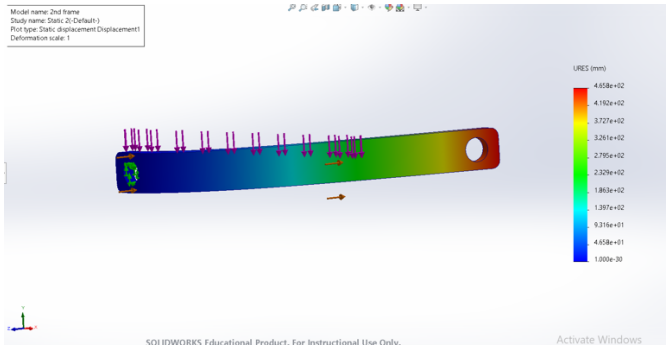


Fig 4.3: Displacement (Front Bow)

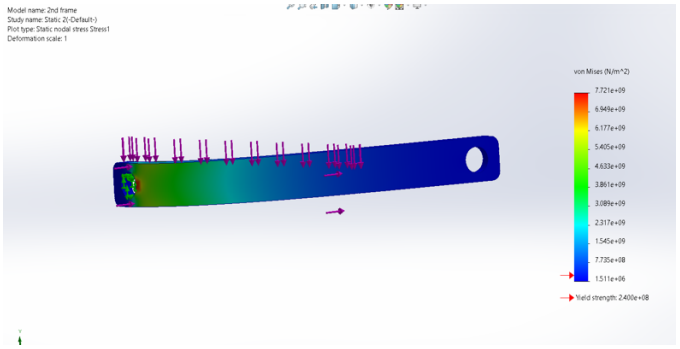


Fig 4.4: Stress (Front Bow)

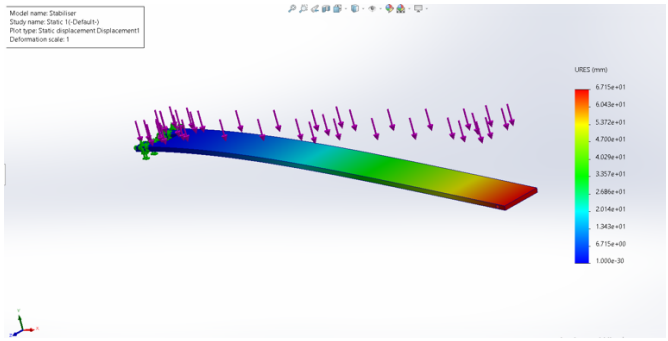


Fig 4.5: Displacement (Stabiliser)

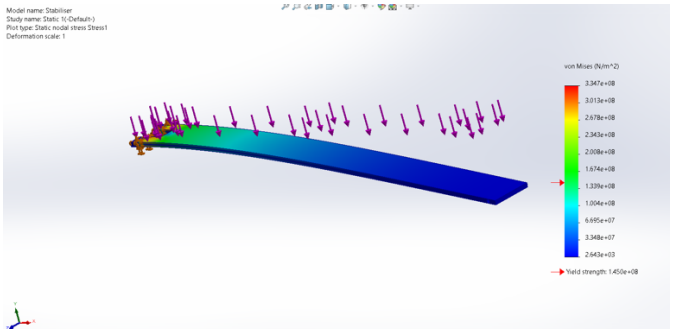


Fig 4.6: Stress (Stabiliser)

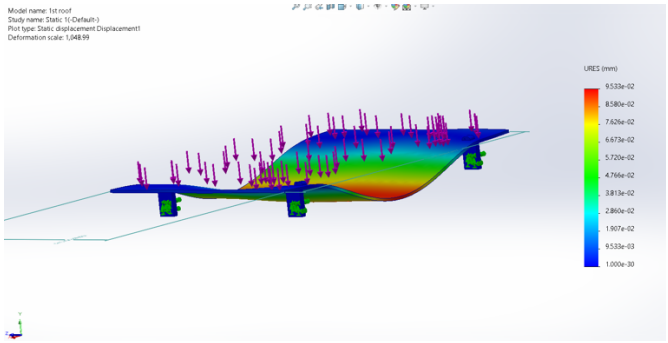


Fig 4.7: Displacement (1st Roof)

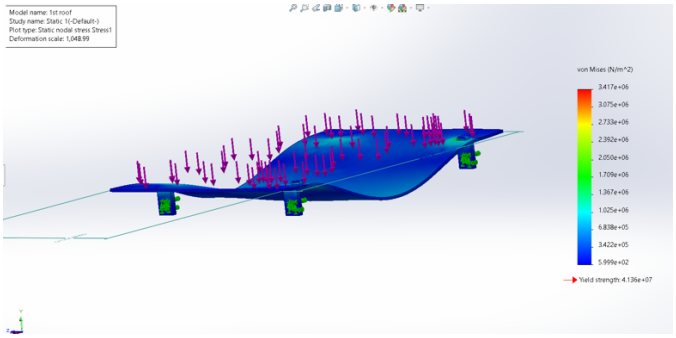


Fig 4.8: Stress (1st Roof)

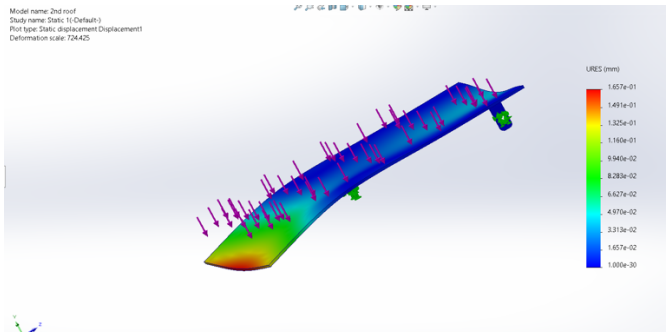


Fig 4.9: Displacement (2nd Roof)

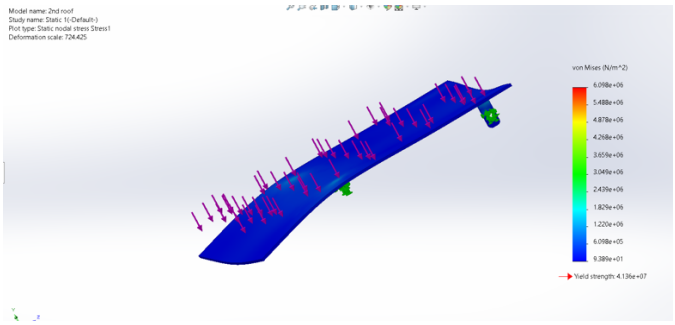


Fig 4.10: Stress (2nd Roof)

Components	Displacement (mm) [From 1.000e-30 to -]	Von Mises Stress (N/m^2)	Yield Strength (N/m^2)
Rear Bow	1.026e+03	3.779e+05	5.515e+07
Front Bow	4.658e+02	5.000e+07	2.400e+08
Stabiliser	6.715e+01	1.395e+08	1.450e+08
1st Roof	9.533e+02	5.999e+02	4.136e+07
2nd Roof	1.657e-01	9.389e+01	4.136e+07

Table 4.3: Stress Analysis Results of Each Awning Structure

In Stress Analysis, according to Table 4.3, all the Von Mises Stress experienced by every component of the structure is under the respective yield strength. Therefore, that explains that the materials used for each component are suitable for their respective purposes. Furthermore, Table 4.3 shows that the stabiliser has the highest Von Mises Stress present of 1.395e+08 N/m^2 as it is very close to reaching its yield strength of 1.450e+08 N/m^2 with the percentage difference of only **3.87%**. On the other hand, 2nd Roof panel has the lowest Von Mises Stress present of 9.389e+01 N/m^2 while the yield strength for that structure is 4.136e+07 N/m^2. This is because higher yield strength material is used for the roof panels with the consideration of other external force such as wind force that can be blown in different directions.

Chapter 5: Conclusion and Recommendations

5.1 Conclusion

All the objectives have been achieved through different processes mainly by Chapter 3 and Chapter 4. The literature review in Chapter 2 closely analyses on the findings of the chosen research and the gaps that were missed to study by those papers. This allows this project to focus on those gaps more which also leads to design and develop design concepts. Thus, Objective 1 has been achieved. Next, Objective 2 has also been achieved through literature review by carefully learning on the nature of awning from Lee, et al (2018) and the use of rain sensor for the structural operation from Muhammad, et al (2023) and use that to develop smart awning system. The test results in Table 4.1 and 4.2 shows that the system is capable to response the rain in short amount of time with the operating time taken as minimum as possible which completed Objective 3. At the same time, Objective 4 is also achieved by the assembly of the small-scale model that can show case the key concept of the prototype which is to detect rain and response. Trouble shooting process was mostly present during the assembly and the programming where it required to re model the entire structure again due to unfit or unsuitable alignment of the components and debug and refine the code to run smoothly without any error. Finally, the prototype could run without any issues, and this can be seen monitored in Fig 3.28 and 3.29. Therefore, Objective 5 has also been achieved. In conclusion, the project aim has also been achieved.

5.2 Recommendations

The IoT cloud-based control and monitoring done by Muhammad, et al (2023) is yet to researched and implemented inside this developed system. That would open path to more options in terms of user interface as it allows the user to monitor from anywhere and get notified in certain conditions. Next, it is also the future advancement to integrate the system with the wind sensor because the installation of wind sensor would increase the safety of the operation of the awning since it could detect strong winds and give alarm to the user that the awning cannot and cannot be operated during the serious weathers.

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